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Uncovering Hierarchical Structure in LLM Embeddings Using Geometric and Topological Measures

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Abstract: The rapid advancement of large language models (LLMs) has driven major advances across scientific domains, yet the hierarchical, geometric, and topological structure of their embedding spaces remains poorly understood. In this work, we introduce a comprehensive framework for evaluating LLM embeddings through six complementary metrics that characterize hierarchy, geometry, and topology. Specifically, δ-hyperbolicity and ultrametricity quantify the degree of tree-like structure, while Neighbor Joining evaluates how well pairwise distances support hierarchical reconstruction. The Ollivier-Ricci curvature captures local geometric behavior, persistent

homology measures multiscale topological stability, and fractal dimension estimates intrinsic complexity. Applying this framework to embeddings from multiple protein LLMs, we reveal how different models encode hierarchical relationships and structural organization. Evaluating the structure of these embeddings in the representation space helps improve geometry-aware model design. This is critical in domains such as bioinformatics and healthcare, where real-world hierarchical and relational structures matter. Our results demonstrate that geometric and topological analysis provides a guiding lens for understanding and potentially improving the representational quality of LLM embeddings.

1 Introduction

The integration of large language models (LLMs) in healthcare has advanced rapidly. These models generate embeddings that capture rich contextual and sequential information, enabling downstream tasks such as protein function prediction and structure inference. This mapping of proteins in hyperbolic space Dress enables efficient clustering of functionally similar proteins, thus supporting applications in Functional Annotation of Proteins Ali et al.. However, understanding the geometric properties of the embeddings is crucial to improving reliability Nickel and Kiela. One way to assess the structural properties of embeddings is through δ-hyperbolicity, a concept introduced by Gromov Gromov. δ-hyperbolicity quantifies the extent to which a metric space deviates from being an ideal tree Yang et al. like structure, where lower δ values indicate structures closer to a tree-like geometry, which suggests hierarchical relationships among data points. In contrast, a higher δ indicates a more complex, non-hierarchical structure. A major empirical study tested whether social and

information networks are tree-like using Gromov's δ-hyperbolicity Adcock et al., showing that although traditional metrics miss such structure, hyperbolic analysis reveals meaningful hierarchical patterns. δ-hyperbolicity quantifies approximate tree-likeness in complex biological networks, simplifying tasks such as distance estimation, network classification, and core–periphery identification Abu-Ata and Dragan. Ultrametricity Chierchia and Perret is another popular stricter metric for capturing hierarchical structure, ideal for deterministic, nested data such as phylogenetic trees. In techniques like UPGMA (Unweighted Pair Group Method with Arithmetic Mean) and WPGMA (Weighted Pair Group with Arithmetic Mean), which are used to build ultrametric trees in hierarchical clustering, the idea of ultrametricity is fundamental Dubes. Whereas Q-matrix statistics from Neighbor Joining offer a quantitative, algorithmic evaluation of tree-likeness that complements these geometric metrics. Strong adherence to additive tree structure is indicated by low normalized Q values Gascuel and Steel; Atteson. NJ evaluates whether pairwise distances are compatible with an additive tree by transforming the distance matrix into a Q-matrix, demonstrating how strongly each pair of points supports hierarchical merging.

Beyond tree-likeness, embedding geometry can exhibit additional structural properties that strictly hierarchical metrics do not capture. We incorporate three complementary geometric and topological measures that probe different, orthogonal aspects of the space. These include Ollivier–Ricci curvature Ollivier; Ni et al. for local geometric behavior, persistent homology Edelsbrunner et al. for multiscale topological structure, and fractal dimension Grassberger and Procaccia for intrinsic geometric complexity.

We adapt δ-hyperbolicity, ultrametricity, neighborhood joining, and three complementary geometric and topological measures to the domain of genomic sequencing data analyses by evaluating the embeddings generated by LLMs

such as "Tasks Assessing Protein Embeddings (TAPE)" Rao et al., ESM2 Lin et al., Seqvec Heinzinger et al., and ProtT5 Elnaggar et al.. We aim to provide a deeper understanding of the geometric properties underlying LLM-generated representations. The key contributions of this study are as follows:

1. We generate embeddings for three datasets using different LLMs and evaluate them with six complementary geometric and topological metrics spanning hierarchy, local geometry, topology, and intrinsic complexity.

2. We provide a comparative geometric analysis of the embeddings of several protein LLMs (SeqVec, ESM2, TAPE, ProtT5), showing how the model design shapes the embedding structure.

3. We assess embedding quality through classification and clustering, demonstrating that geometric properties correlate with downstream performance.

2 Related Work

LLMs have become widely used to produce rich embeddings that boost performance across diverse tasks. Similar models are now used in protein biology to map sequence and structure for better function prediction Ali et al.. δ-hyperbolicity, introduced by Gromov Gromov, provides a geometric measure of how tree-like a metric space is. It has been widely applied in graph theory and computational geometry Krioukov et al. to analyze structural properties relevant to computational problems. Authors in Chami et al. show that HypHC closely approximates the optimal tree and outperforms other clustering methods. Authors in Chierchia and Perret frame hierarchical clustering as an optimization problem, seeking the ultrametric that best fits the data.

Recent work shows that geometric properties of embeddings, including hyperbolicity, can inform clustering and classification Yang et al., but these ideas

have not been applied to biological sequence embeddings. Since many biological processes follow tree-like evolutionary patterns Felsenstein; Abu-Ata and Dragan, δ-hyperbolicity provides a natural tool for assessing and improving models in phylogenetics, viral evolution, and epidemiology. Hyperbolic embedding techniques have also shown benefits in drug–target prediction and functional protein annotation Dress; Ali et al., highlighting the broader potential of hyperbolic geometry in biological modeling. Recent studies use geometric tools such as Ricci curvature, persistent homology, and fractal dimension to analyze neural representations Ni et al.; Carlsson; Facco et al.. We incorporate these ideas to provide a unified geometric and topological assessment of protein LLM embeddings.

3 Proposed Approach

In this section, we outline our framework for evaluating embeddings generated by large language models (LLMs). Our analysis proceeds as follows :

1. Generate embeddings using the different LLMs.

2. Compute Euclidean and Poincaré distance matrices.

3. Evaluate tree-like structure using three complementary metrics: δ-hyperbolicity, ultrametricity, and Neighbor Joining (via the distribution of Q-matrix).

4. Assess additional geometric and topological structure using three complementary metrics.

5. Analyze the classification and clustering results to understand the geometric properties of the embeddings.

3.1 LLMs for embedding generation

We employ four LLMs, including SeqVec, ESM-2, TAPE, and ProtT5, to generate embeddings and evaluate how well they capture structural patterns in biological sequences.

Seqvec Heinzinger et al. SeqVec is an ELMo-inspired protein language model that learns contextual embeddings directly from large unlabeled protein datasets such as UniRef50 Suzek et al.. It generates fast, alignment-free representations that capture meaningful sequence features and can rival traditional evolutionary methods Sarzynska-Wawer and Wawer.

ESM2 Lin et al. ESM-2 is a large transformer-based protein language model Lin et al. that predicts structural features directly from single sequences, eliminating the need for MSAs. Scaled to billions of parameters, it captures detailed structural signals Rao et al. and powers fast, high-accuracy tools like ESMFold, advancing single-sequence structure prediction.

TAPE Rao et al. TAPE is a protein representation learning framework that trains transformer, LSTM, and ResNet models on large unlabeled datasets like UniRef50 using self-supervised objectives. It produces contextual embeddings for each sequence and provides the TAPE benchmark, which standardizes downstream tasks to consistently evaluate how well these representations capture structural and functional properties of proteins.

ProtT5 Elnaggar et al. ProtT5 adapts the T5 transformer to protein sequences, training with a BERT-style masked language modeling objective and using only the encoder for efficient inference Elnaggar et al.. Without requiring evolutionary data or MSAs, its embeddings capture significant biophysical

characteristics and perform well on tasks such as secondary structure and localization prediction.

3.2 Distance Matrix Calculation

We compute pairwise distance matrices for all embeddings using both Euclidean and Poincaré metrics to obtain $N \times N$ distance matrices, where N is the total number of protein sequences. Let X be the set of embeddings.

For the Euclidean metric, the distance between two embeddings x_i and x_j is given by:

$$d_E(x_i, x_j) = \sqrt{\sum_{k=1}^{d} (x_{i,k} - x_{j,k})^2}, \tag{1}$$

where d is the embedding dimensionality.

For the Poincaré metric Nickel and Kiela, the hyperbolic distance between two sequence embeddings x_i and x_j within the unit ball is computed as:

$$d_P(x_i, x_j) = \operatorname{arcosh}\left(1 + 2\frac{\|x_i - x_j\|^2}{(1 - \|x_i\|^2)(1 - \|x_j\|^2)}\right). \tag{2}$$

Both metrics capture complementary geometric properties. Euclidean measures standard linear distances. Poincaré captures hierarchical, tree-like relationships in hyperbolic space.

4 Geometric and Topological Measures

We evaluate embedding structure using six metrics spanning hierarchy, geometry, topology, and complexity: δ-hyperbolicity, ultrametricity, Neighbor-Joining, Ollivier–Ricci curvature, persistent homology, and fractal dimension.

4.1 δ-hyperbolicity

The proposed algorithm for evaluating δ-hyperbolicity is outlined in Algorithm 1 and is implemented to assess the geometric structure of LLM-derived protein sequence embeddings. The algorithm begins by taking as input a set of embeddings X computed from different LLMs, a distance function d, and a user-defined number of samples to evaluate. The goal is to calculate the δ-hyperbolicity value δ for the metric space defined by the embeddings. Initially, the embeddings are flattened and padded to a uniform length, then optionally reduced in dimensionality via PCA to retain 99% of variance, ensuring efficient computation. First, it computes the pairwise distance matrix D using the distance function d over the embeddings, where each entry in D represents the distance between a pair of embeddings. The core of the algorithm involves sampling a specified number of random quadruples (a, b, c, w) from the embedding space, rather than iterating exhaustively over all combinations, which helps it to scale to large datasets. To evaluate δ-hyperbolicity, we need to compute the Gromov products for four points a, b, c, and w. The Gromov product $[a, b]_w$ is defined as (also mentioned in Yang et al.):

$$[a, b]_w = \frac{1}{2} \left(d(a, w) + d(b, w) - d(a, b) \right) \tag{3}$$

Where d is the Euclidean or Poincaré distance between embeddings.

The δ-hyperbolicity is then calculated by checking if the following inequality holds for all sampled quadruples a, b, c, and w Yang et al.:

$$[a, c]_w \geq \min([a, b]_w, [b, c]_w) - \delta \tag{4}$$

We estimate δ-hyperbolicity by computing the minimum value of δ that satisfies the above condition for all quadruples in the space.

To estimate the δ-hyperbolicity of LLM-derived embeddings, we sample ran-

dom quadruples of points in the embedding space and compute how much each quadruple deviates from the ideal tree-like geometry. After flattening and optionally reducing the embeddings with PCA, we compute pairwise distances and evaluate the Gromov products for each quadruple. The slack between these products gives a local estimate of the hyperbolic deviation. Aggregating these values across all samples yields the maximum, mean, and standard deviation of δ, which together quantify how tree-like the embedding space is, where lower values indicate stronger hierarchical structure. The algorithm for computing δ Hyperbolicity is discussed in detail in the Supplementary Section.

Using parallel processing, these δ values are calculated for all quadruples sampled, after which the algorithm returns the maximum, average, and standard deviation of δ in all samples. These values collectively quantify the degree of hyperbolicity in the embedding space. A lower average δ indicates a more tree-like space, suggesting that the embedding structure may support hierarchical relationships.

4.2 Ultrametricity

A metric space is said to be **ultrametric** if the distance function d satisfies the *ultrametric inequality*, also called a stricter version of the triangle inequality Chierchia and Perret:

$$d(i,k) \leq \max\{d(i,j), d(j,k)\} \quad \forall i,j,k \tag{5}$$

This is a stronger condition than the standard triangle inequality. It enforces a strict hierarchy, making it ideal for modeling perfect tree structures like phylogenetic trees and hierarchical clustering dendrograms.

The algorithm 2 determines whether a given distance matrix $D \in \mathbb{R}^{n\times n}$ satisfies the ultrametric inequality. Specifically, the ultrametric condition requires

Algorithm 1: Compute δ-Hyperbolicity

Input: Set of embeddings X, distance function d, number of sampled quadruples S

Output: Estimated δ-hyperbolicity values: $\delta_{\max}, \delta_{\text{avg}}, \delta_{\text{std}}$

1: Sample S random quadruples (a, b, c, w) from X

2: Initialize empty list $\Delta \leftarrow [\,]$

3: **for** each quadruple (a, b, c, w) **do**

4: Compute Gromov products: {using Eq 3 and Eq 4}

$$[a, b]_w = \frac{1}{2}\left(d(a, w) + d(b, w) - d(a, b)\right)$$

$$[b, c]_w = \frac{1}{2}\left(d(b, w) + d(c, w) - d(b, c)\right)$$

$$[a, c]_w = \frac{1}{2}\left(d(a, w) + d(c, w) - d(a, c)\right)$$

5: Compute temporary slack:

$$\delta_{\text{temp}} = [a, c]_w - \min\left([a, b]_w, [b, c]_w\right)$$

6: Append δ_{temp} to Δ

7: **end for**

8: Compute:

$$\delta_{\max} \leftarrow \max(\Delta),$$
$$\delta_{\text{avg}} \leftarrow \text{mean}(\Delta),$$
$$\delta_{\text{std}} \leftarrow \text{std}(\Delta)$$

9: **return** $\delta_{\max}, \delta_{\text{avg}}, \delta_{\text{std}}$

that for any triplet of points (i, j, k), the largest pairwise distance among them must not exceed the maximum of the other two distances. To verify this, the algorithm iterates over all unique sampled combinations of triplets in the distance matrix. For each triplet, it retrieves the corresponding distances d_{ij}, d_{ik}, and d_{jk}, then identifies the largest of the three. It compares this maximum value to the sum of the other two (smaller) distances, adjusted by a small tolerance ϵ to account for numerical imprecision. If the maximum distance exceeds this sum by more than ϵ, the matrix is deemed to violate the ultrametric property, and the algorithm returns **False** immediately. If all triplets conform to the inequality, it concludes by returning **True**, confirming that the entire matrix exhibits ultrametricity. This algorithm is particularly valuable in fields such as phylo-

genetics and hierarchical clustering, where validating the ultrametric nature of data helps ensure the reliability of tree-based representations.

For each triplet, we check whether the largest of the three distances exceeds the other two by more than a small numerical tolerance. A violation is decided on the basis of it. We quantify the degree to which the distance matrix resembles a completely hierarchical ultrametric space by adding the maximum, mean, and standard deviation of these violations across all samples. As anticipated in applications such as phylogenetics and hierarchical clustering, this offers a straightforward method to assess whether the embeddings represent nested or tree-like relationships.

Algorithm 2: Ultrametricity of a Distance Matrix

Input: Distance matrix $D \in \mathbb{R}^{n \times n}$, sample size S, tolerance ϵ

Output: Dictionary containing ultrametricity violation metrics

Initialize counters: num_violations $\leftarrow 0$, total_triples $\leftarrow 0$, and empty list violations $\leftarrow []$

Randomly sample S unique triples $\{(i, j, k)\}$ from $\{1, \ldots, n\}$ without replacement

for each sampled triple (i, j, k) **do**

  Extract distances $d_{ij} = D[i, j]$, $d_{jk} = D[j, k]$, and $d_{ik} = D[i, k]$

  Compute violation terms:

$$v_1 = d_{ik} - \max(d_{ij}, d_{jk})$$

$$v_2 = d_{ij} - \max(d_{ik}, d_{jk})$$

$$v_3 = d_{jk} - \max(d_{ij}, d_{ik})$$

  Compute violation $\nu = \max(v_1, v_2, v_3, 0)$

  if $\nu > \epsilon$ **then**

    Append ν to violations

    Increment num_violations $\leftarrow$ num_violations $+ 1$

  end if

  Increment total_triples $\leftarrow$ total_triples $+ 1$

end for

// Compute statistics over violations:

max_violation = max(violations) if violations $\neq \varnothing$ else 0

avg_violation = mean(violations) if violations $\neq \varnothing$ else 0

std_violation = std(violations) if violations $\neq \varnothing$ else 0

return max_violation, avg_violation, std_violation

4.3 Neighbor Joining

We use another metric called Neighbor Joining (NJ) to gauge how closely the pairwise distance structure of the embeddings follows a tree-like geometry. We employed a Neighbor Joining (NJ)-based analysis produced from the NJ Q-matrix Trees.

Given an $n \times n$ distance matrix D, the NJ algorithm defines the Q-matrix as

$$Q(i,j) = (n-2)D(i,j) - \sum_{k=1}^{n} D(i,k) - \sum_{k=1}^{n} D(j,k) \tag{6}$$

where pairs with the most negative $Q(i,j)$ values are preferentially joined during tree construction. Rather than explicitly reconstructing a phylogenetic tree, we use the NJ Q-matrix, which transforms the distance matrix into a form that highlights how strongly each pair of points supports hierarchical merging. Pairs with highly negative Q-values are those most consistent with additive tree geometry. This makes the Q-matrix a useful diagnostic tool for assessing tree-likeness directly from distances.

Algorithm 3 shows the computation that uses the Neighbor-Joining (NJ) Q-matrix to compute tree-likeness statistics from a given distance matrix. By modifying pairwise distances based on their global relationships, it first creates the Q-matrix. After that, all distinct pairwise Q-values are extracted, converted to absolute values, and their distribution is summarized using the maximum, mean, and standard deviation. The degree to which the data exhibit a hierarchical or tree-like structure is concisely captured by these statistics.

To obtain a global measure of hierarchical consistency, we summarize the distribution of all off-diagonal Q-values by reporting their maximum, mean, and standard deviation after normalization. The lower mean and maximum values indicate that the distance matrix more closely follows the NJ criterion and better approximates a tree metric. Combined with other geometric indicators such

as hyperbolicity and curvature, these scores provide a robust measure of how strongly an embedding reflects hierarchical structure. Rather than constructing the full tree, we summarize the distribution of Q-matrix values to capture the embedding's overall consistency with a tree-structured geometry.

Algorithm 3: Neighbor Joining–Based Tree-Likeness Scoring

Input: Distance matrix $D \in \mathbb{R}^{n \times n}$
Output: Dictionary containing NJ score statistics
Let $n \leftarrow$ number of samples in D
if $n < 3$ **then**
  return nj_max $= 0$, nj_avg $= 0$, nj_std $= 0$
end if
Compute the Neighbor Joining Q-matrix $Q \in \mathbb{R}^{n \times n}$:

$$Q(i,j) = (n-2)D(i,j) - \sum_{k=1}^{n} D(i,k) - \sum_{k=1}^{n} D(j,k)$$

Initialize empty list $\mathcal{Q} \leftarrow [\,]$
for $i = 1$ to n **do**
  for $j = i + 1$ to n **do**
    Append $Q(i,j)$ to $\mathcal{Q}$
  end for
end for
Normalize Q-values: $\mathcal{Q} \leftarrow \{\, |q| \mid q \in \mathcal{Q} \,\}$
nj_max $\leftarrow \max(\mathcal{Q})$
nj_avg $\leftarrow \text{mean}(\mathcal{Q})$
nj_std $\leftarrow \text{std}(\mathcal{Q})$
return nj_max, nj_avg, nj_std

4.4 Ollivier-Ricci Curvature

To capture local geometric properties complementary to global hyperbolicity, we computed the discrete Ollivier-Ricci curvature Ollivier on the k-nearest neighbor graph constructed from embeddings (see Algorithm 4). For adjacent points x_i and x_j, the Ollivier-Ricci curvature is defined as:

$$\kappa(x_i, x_j) = 1 - \frac{W_1(\mu_i, \mu_j)}{d(x_i, x_j)} \tag{7}$$

where $W_1(\mu_i, \mu_j)$ is the Wasserstein-1 distance between probability distributions μ_i and μ_j centered at x_i and x_j respectively, typically defined as uniform distributions over their k-nearest neighbors Ni et al.. Negative curvature ($\kappa < 0$) indicates the local exponential divergence characteristic of hyperbolic geometry and tree-like structure Sandhu et al., while positive curvature suggests clustering or community structure. We report the mean, standard deviation, minimum, and proportion of edges with negative curvature in the k-NN graph.

Algorithm 4: Compute Ollivier-Ricci Curvature

Input: Embedding matrix $X \in \mathbb{R}^{n \times d}$, number of neighbors k

Output: Mean Ricci curvature $\bar{\kappa}$, standard deviation σ_κ, minimum κ_{min}, negative ratio r_{neg}

Construct k-NN graph $G = (V, E)$ from X
Initialize $\kappa_values \leftarrow []$
for each edge $(i, j) \in E$ **do**
    $N_i \leftarrow$ k-nearest neighbors of x_i
    $N_j \leftarrow$ k-nearest neighbors of x_j
    Define uniform distributions: $\mu_i = \text{Uniform}(N_i)$, $\mu_j = \text{Uniform}(N_j)$
    Compute Wasserstein-1 distance: $W_1 = W_1(\mu_i, \mu_j)$
    $\kappa(i, j) \leftarrow 1 - \frac{W_1}{d(x_i, x_j)}$
    Append $\kappa(i, j)$ to κ_values
end for
$\bar{\kappa} \leftarrow \text{mean}(\kappa_values)$
$\sigma_\kappa \leftarrow \text{std}(\kappa_values)$
$\kappa_{min} \leftarrow \min(\kappa_values)$
$r_{neg} \leftarrow \frac{\#\{\kappa < 0\}}{|\kappa_values|}$
return $\bar{\kappa}, \sigma_\kappa, \kappa_{min}, r_{neg}$

4.5 Persistent homology

We employ 0-dimensional persistent homology Edelsbrunner et al.; Carlsson to quantify the topological structure of embeddings across scales (see Algorithm 5). Persistent homology tracks connected components as we increase the distance threshold ϵ from 0 to ∞. Each connected component has a "birth" time (when it first appears) and "death" time (when it merges with another component).

The persistence lifetime $\ell = \text{death} - \text{birth}$ measures the stability of that topological feature Chazal et al..

For tree-like structures with clear hierarchical organization, components merge rapidly, yielding short persistence lifetimes. In contrast, flat or diffuse geometries exhibit long persistence lifetimes as components remain separated across many scales. We compute the persistence diagram using the Ripser algorithm Bauer and report:

- Mean persistence lifetime: $\bar{\ell} = \frac{1}{n-1} \sum_{i=1}^{n-1} \ell_i$
- Maximum persistence lifetime: $\max_i \ell_i$
- Persistence entropy: $H = -\sum_{i=1}^{n-1} p_i \log p_i$ where $p_i = \ell_i / \sum_j \ell_j$

Lower persistence values indicate stronger hierarchical consolidation.

Algorithm 5: Compute Persistent Homology Features

Input: Distance matrix $D \in \mathbb{R}^{n \times n}$
Output: Mean persistence $\bar{\ell}$, max persistence ℓ_{max}, persistence entropy H

Compute 0-dimensional persistence diagram using Ripser: $\{(\text{birth}_i, \text{death}_i)\}_{i=1}^{n-1}$
Compute lifetimes: $\ell_i = \text{death}_i - \text{birth}_i$
$\bar{\ell} \leftarrow \frac{1}{n-1} \sum_{i=1}^{n-1} \ell_i$
$\ell_{max} \leftarrow \max_i \ell_i$
Normalize: $p_i \leftarrow \frac{\ell_i}{\sum_{j=1}^{n-1} \ell_j}$
$H \leftarrow -\sum_{i=1}^{n-1} p_i \log p_i$
return $\bar{\ell}, \ell_{max}, H$

4.6 Fractal Dimension

We estimate intrinsic dimensionality using the correlation dimension Grassberger and Procaccia, derived from the scaling behavior of the correlation integral for points x_i. Details are mentioned in Algorithm 6. Correlation integral

for points x_i can be defined by:

$$C(r) = \lim_{n\to\infty} \frac{1}{n^2} \sum_{i\neq j} \mathbb{I}(d(x_i, x_j) < r) \tag{8}$$

In the scaling region, $C(r) \propto r^{d_f}$, so d_f can be estimated from the slope of $\log C(r)$ vs $\log r$ Camastra and Staiano. We use the two-nearest-neighbor method Facco et al. for robust estimation:

$$d_f = \left[\frac{1}{n} \sum_{i=1}^{n} \log \left(\frac{r_2(x_i)}{r_1(x_i)} \right) \right]^{-1} \tag{9}$$

where $r_1(x_i)$ and $r_2(x_i)$ are the distances to the first and second nearest neighbors of x_i. Lower fractal dimension signals embeddings concentration near low-dimensional manifolds, whereas higher values indicate a more diffuse, isotropic spread in high-dimensional space Levina and Bickel.

Algorithm 6: Estimate Fractal Dimension

Input: Embedding matrix $X \in \mathbb{R}^{n\times d}$
Output: Mean fractal dimension $\bar{d}_f$, standard deviation σ_{d_f}, median $\tilde{d}_f$

$d_f_values \leftarrow []$
for each point $x_i \in X$ **do**
  Find distances to 2 nearest neighbors: $r_1(x_i), r_2(x_i)$
  $d_{f,i} \leftarrow \left[\log \left(\frac{r_2(x_i)}{r_1(x_i)} \right) \right]^{-1}$
  Append $d_{f,i}$ to d_f_values
end for
$\bar{d}_f \leftarrow \text{mean}(d_f_values)$
$\sigma_{d_f} \leftarrow \text{std}(d_f_values)$
$\tilde{d}_f \leftarrow \text{median}(d_f_values)$
return $\bar{d}_f, \sigma_{d_f}, \tilde{d}_f$

5 Experimental Setup

We performed the experiments on a computer running 64-bit Windows 10 with an Intel(R) Core i5 processor at 2.10 GHz and 32 GB of RAM. For experi-

ments, we randomly split the data into 60-10-30% for training, validation, and testing. We repeat the experiments 5 times and report the average and standard deviation. For the sake of reproducibility, we have made code available online [1].

5.1 Data Statistics

We use 3 real-world protein sequence datasets. The summary of the datasets used for the experimentation is given below:

5.1.1 PDB186

We use a benchmark dataset from the literature (PDB186) for predicting DNA-protein binding, a binary classification task Lou et al.. More formally, the dataset contains DNA-binding and non-binding protein sequences. The research task is to predict whether the DNA and protein sequences will bind. The minimum, maximum, and average sequence length in the data are 64, 1323, and 264.693, respectively. The total number of sequences is 186, of which 93 bind and 93 do not.

5.1.2 Breast Cancer

Our Membranolytic anticancer peptides (ACPs) dataset Grisoni et al. contains 949 protein sequences annotated for anticancer activity against breast cancer cell lines. The target labels are categorized into "very active", "moderately active", "experimental inactive", and "virtual inactive" groups. Table 1 shows the distribution in detail.

5.1.3 Lung Cancer

The Membranolytic anticancer peptides (ACPs) dataset Grisoni et al. contains 901 sequences annotated for anticancer activity against Lung cancer cell lines.

[1] `https://github.com/pchourasia1/Hyperbolicity_LLM_Embeddings`

ACPs Category	Count	Min.	Max.	Average
Inactive-Virtual	750	8	30	16.64
Moderate Active	98	10	38	18.44
Inactive-Experimental	83	5	38	15.02
Very Active	18	13	28	19.33
Total	949	-	-	-

Table 1: Dataset statistics for the Breast cancer data. Columns represent the minimum, maximum, and average sequence lengths.

ACPs Category	Count	Min.	Max.	Average
Inactive-Virtual	750	8	30	16.64
Moderate Active	75	11	38	17.76
Inactive-Experimental	52	5	38	14.5
Very Active	24	13	28	20.70
Total	901	-	-	-

Table 2: Dataset statistics for the Lung cancer data. Columns represent the min., max., and average lengths of the sequence.

The target labels are classified into four groups: "very active," "moderately active," "experimental inactive," and "virtual inactive" groups. Table 2 shows the distribution in detail.

5.2 Classification

For the classification task, we use classifiers like SVM, Naive Bayes, MLP, KNN, Random Forest, Logistic Regression, and Decision Tree. Evaluation metrics include accuracy, precision, recall, weighted F1, macro F1, ROC-AUC, and training runtime.

5.3 Clustering

For the clustering task, we use k-means Lloyd, k-modes Huang, and the agglomerative clustering algorithm Ward Jr. The evaluation metrics used were the Silhouette Coefficient Rousseeuw, Calinski-Harabasz Score Calinski and Harabasz,

and the Davies-Bouldin Score Davies and Bouldin.

5.3.1 Algorithms

k-means Lloyd The k-means Lloyd clusters objects based on the Euclidean mean. The k-means clustering partitions the data into K groups by repeatedly assigning points to the nearest centroid and updating the centroids as the mean of their assigned points. This continues until the assignments stabilize, resulting in compact, well-separated clusters Kanungo et al..

k-modes Huang We also perform clustering with k-modes Huang, which is a variant of k-means using *modes* instead of *means*. Here, pairs of objects are subject to a dissimilarity measure (*e.g.*, Hamming distance) rather than to the Euclidean mean.

Agglomerative Ward Jr All agglomerative hierarchical clustering methods start from a proximity matrix of either similarities or distances between elements Nei. Agglomerative clustering is a bottom-up algorithm that starts with each data point as its own cluster. It repeatedly merges the closest pairs of clusters based on a chosen distance metric until all points are grouped into a single cluster or a set number of clusters Mojena.

5.3.2 Evaluation Metric

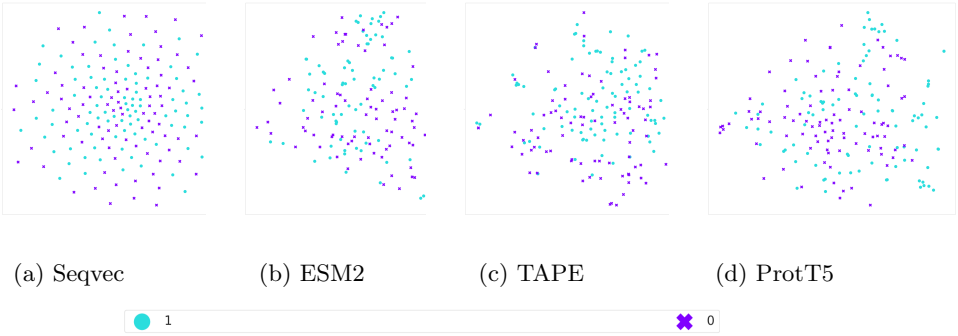
Silhouette Coefficient Rousseeuw Measures how similar a point is to its own cluster versus similarity with other clusters. Scores range from -1 to 1, with higher values indicating better clustering.

Calinski–Harabasz Score Calinski and Harabasz Evaluates clustering quality using between-cluster and within-cluster dispersion. Higher values indicate more distinct, well-separated clusters.

Davies–Bouldin Score Davies and Bouldin Computes the ratio of within-cluster to between-cluster distances. Lower values indicate better separation between clusters.

5.4 Visualization

A popular visualization technique, named t-SNE Van der M. and Hinton, is employed to visualize the feature vectors generated by the SeqVec Heinzinger et al. method. The t-SNE plots are shown in Figure 1 for PDB186, Figure 2 for Lung Cancer, and Figure 3 for breast cancer datasets. The main idea for reporting the t-SNE plots is to assess whether there is a clear class separation between the labels in the datasets using embeddings generated by the respective LLMs. We observe that clusters overlap for Seqvec in the Lung cancer and breast cancer datasets, indicating that no clear decision boundary exists in the data.

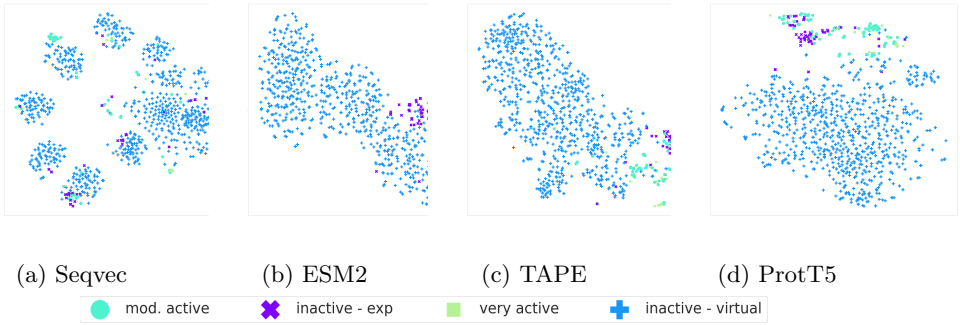


(a) Seqvec (b) ESM2 (c) TAPE (d) ProtT5

Figure 1: t-SNE visualization of different embeddings from **PDB 186 dataset**. The Figure is best seen in color.

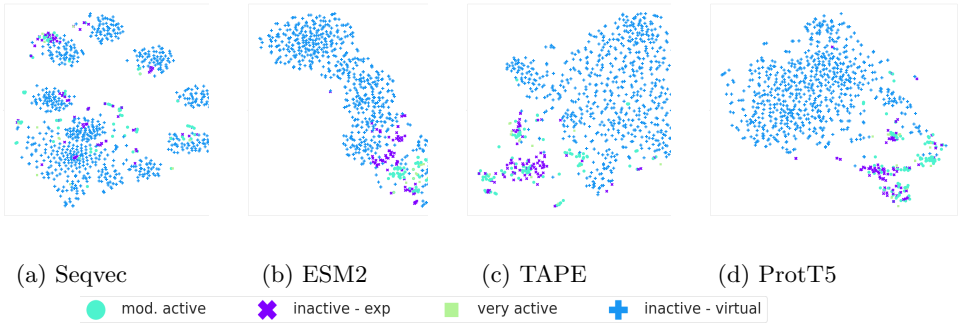
6 Results And Discussion

In the first step, we simulate the data to understand the hyperbolicity values for known structures using the following synthetic data.



(a) Seqvec (b) ESM2 (c) TAPE (d) ProtT5

Figure 2: t-SNE visualization of different embeddings from **Lung Cancer dataset**. The Figure is best seen in color.



(a) Seqvec (b) ESM2 (c) TAPE (d) ProtT5

Figure 3: t-SNE visualization of different embeddings from **Breast Cancer dataset**. The Figure is best seen in color.

Sphere Space Data Computation To simulate data in the sphere space, we generate n random points on the surface of a unit sphere in a high-dimensional space. Each point is sampled from a standard normal distribution in $\mathbb{R}^d$, and subsequently normalized to lie on the unit sphere by dividing each point by its Euclidean norm. The distance matrix for these points is computed using the Euclidean distance metric. The δ-hyperbolicity values for this space are then calculated by evaluating the shortest path distances between all possible pairs of points, followed by applying the δ-hyperbolicity calculation method described in Section 3.

Dense Graph Data Computation A dense graph is generated using the Erdős-Rényi model, where n nodes are connected with a high probability, $p = 0.8$. The shortest path distances are computed using the Floyd–Warshall algorithm. δ-hyperbolicity is evaluated from this distance matrix, assigning a large finite value to any disconnected pairs to keep the computation valid.

Poincaré Space Data Computation To simulate data in the Poincaré space, we sample n random points in the two-dimensional Poincaré ball by drawing radial distances $r \in [0, 1)$ and angles $\theta \in [0, 2\pi)$, then converting to Cartesian coordinates via $x = r\cos\theta$ and $y = r\sin\theta$. The geodesic distances of pairs are calculated using the hyperbolic distance formula, which incorporates both the Euclidean separation and the distance of each point from the origin.

Metric Space	δ
Sphere Space - δ:	0.32 ± 0.19
Dense Graph - δ:	0.31 ± 0.25
Poincaré Space - δ:	0.2867 ± 0.3328

Table 3: Hyperbolicity δ values (average and standard deviation) for synthetically generated data.

The differences in δ-hyperbolicity across the three synthetic spaces reflect their underlying geometry as shown in Table 3. Sphere and dense graph data show moderately higher δ values of 0.32 ± 0.19 and 0.31 ± 0.25, respectively, due to curvature and high connectivity, both of which reduce tree-likeness. The Poincaré space, on the other hand, produces the lowest δ value of 0.2867 ± 0.3328, which is consistent with its inherently hyperbolic, tree-like structure Nickel and Kiela. This illustrates how a space's tree-like appearance is shaped by geometry.

6.1 Hierarchical, Geometric and Topological Results

6.1.1 PDB186 dataset

Table 4 summarizes the geometric structure of the embeddings across all metrics. TAPE has the highest tree-likeness in the Poincaré space, with the lowest fractal dimension, hyperbolicity, ultrametricity, and NJ deviation. SeqVec shows the most negative Ricci curvature and the highest persistence entropy, indicating strong local hyperbolicity but weaker global hierarchy.ProtT5 exhibits significant deviations from tree structure and performs poorly in Poincaré space. With the lowest δ_{avg}, U_{avg}, and NJ values under Euclidean distances, ProtT5 becomes the most tree-like, and ESM performs the worst. In general, Poincaré distances yield far smaller and more stable hierarchy metrics, confirming that hyperbolic geometry captures tree structure more effectively than Euclidean space.

Overall, the results show that (i) ProtT5 works best when utilizing Euclidean distances; (ii) TAPE is the most hierarchical in hyperbolic space; and (iii) SeqVec shows great local hyperbolicity but weaker global hierarchy. Results show that the tree-likeness of the resulting representation space is significantly influenced by both the embedding model and the underlying distance geometry.

Dist Matrix	Embedding	$\boldsymbol{\delta}_{\text{avg}}$	$\boldsymbol{U}_{\text{avg}}$	$\mathbf{NJ}_{\text{avg}}$	$\mathbf{Ricci}_{\text{avg}}$	$\mathbf{PersEnt}_{\text{avg}}$	$\mathbf{FracDim}_{\text{avg}}$
Poincare	SeqVec	0.0110	0.0242	30.5558	**-0.1712**	**19.0043**	29.9867
	ESM	0.0183	0.0124	19.3127	-0.0650	0.3970	inf
	TAPE	**0.0030**	**0.0083**	**16.3208**	-0.0642	7.3302	**16.3534**
	ProtT5	0.0446	0.3038	567.0299	-0.0652	1.1681	19.4370
Euclidean	SeqVec	1.6227	3.6570	7276.7976	-0.1712	19.0043	29.9867
	ESM	0.8727	0.7193	920.0283	-0.0650	0.3970	inf
	TAPE	0.4178	1.0126	2123.9591	-0.0642	7.3302	16.3534
	ProtT5	**0.0418**	**0.1300**	**318.6305**	-0.0652	1.1681	19.4370

Table 4: Comparison of geometric and topological metrics across different embeddings for the PDB186 dataset, grouped by the distance matrix. The log ratio in the fractal dimension formula approaches zero, and its reciprocal explodes to infinity as a result of ESM embeddings placing numerous nearest neighbors at approximately identical distances, which results in the inf value.

6.1.2 Lung Cancer dataset

The Lung Cancer results show that geometric behavior varies strongly across embeddings and distance metrics. As shown in Table 5, in the Poincaré space, ESM2 is the most hierarchical, with the lowest hyperbolicity (0.0016), ultrametricity (0.0022), and NJ deviation (40.29), as well as the most negative Ricci curvature. TAPE shows the lowest fractal dimension (18.57), indicating a more compact manifold, while SeqVec exhibits the highest persistence entropy, reflecting greater multiscale variability. ProtT5 shows weak hierarchy in Poincaré space but becomes the most tree-like under Euclidean distances, with the lowest δ (0.1010), U (0.2413), and NJ deviation (2296.54). Overall, ESM2 and TAPE demonstrate stronger hyperbolic structure, whereas ProtT5 appears more separable in Euclidean geometry.

Dist Matrix	Embedding	$\boldsymbol{\delta}_{\text{avg}}$	$\boldsymbol{U}_{\text{avg}}$	$\mathbf{NJ}_{\text{avg}}$	$\mathbf{Ricci}_{\text{avg}}$	$\mathbf{PersEnt}_{\text{avg}}$	$\mathbf{FracDim}_{\text{avg}}$
Poincare	SeqVec	0.0026	0.0069	118.3239	-0.0637	**22.4044**	32.0288
	ESM	**0.0016**	**0.0022**	**40.2858**	**-0.0849**	20.7693	32.9002
	TAPE	0.0040	0.0084	72.4654	-0.0653	6.4817	**18.5660**
	ProtT5	0.0305	0.0759	715.5995	-0.0621	1.4096	20.9063
Euclidean	SeqVec	0.5904	1.2987	28674.5853	-0.0637	**22.4044**	32.0288
	ESM	1.0897	1.4986	27513.1399	**-0.0849**	20.7693	32.9002
	TAPE	0.5944	1.2251	10738.2480	-0.0653	6.4817	**18.5660**
	ProtT5	**0.1010**	**0.2413**	**2296.5416**	-0.0621	1.4096	20.9063

Table 5: Comparison of geometric and topological metrics across different embeddings for the Lung Cancer dataset, grouped by the distance matrix.

6.1.3 Breast Cancer dataset

The Breast Cancer results in Table 6 show consistent geometric differences across embeddings. In the Poincaré space, ESM2 is the most hierarchical, achieving the lowest hyperbolicity (0.0017), ultrametricity (0.0023), and NJ deviation (42.74), along with the most negative Ricci curvature. Here also, TAPE shows the lowest fractal dimension (18.12), indicating a compact embedding manifold, while SeqVec exhibits the highest persistence entropy (22.23), reflecting more

topological diversity. With significantly higher ultrametricity (0.0758) and NJ values (752.72), ProtT5 exhibits the weakest Poincaré hierarchy. Similar to patterns seen in previous datasets, ProtT5 becomes the most tree-like under Euclidean distances, exhibiting the lowest δ (0.1073), U (0.2458), and NJ deviation (2487.96). Overall, we can conclude here that ESM2 and TAPE exhibit stronger hyperbolic structure, whereas ProtT5 appears more separable in Euclidean space, and SeqVec remains more topologically fragmented.

Dist Matrix	Embedding	$\boldsymbol{\delta}_{\text{avg}}$	$\boldsymbol{U}_{\text{avg}}$	$\textbf{NJ}_{\text{avg}}$	$\textbf{Ricci}_{\text{avg}}$	$\textbf{PersEnt}_{\text{avg}}$	$\textbf{FracDim}_{\text{avg}}$
Poincare	SeqVec	0.0025	0.0073	124.5329	-0.0675	22.2295	30.0684
	ESM	**0.0017**	**0.0023**	**42.7396**	-0.0748	20.4887	31.0122
	TAPE	0.0040	0.0087	76.5285	-0.0690	6.4066	18.1227
	ProtT5	0.0306	0.0758	752.7241	-0.0653	1.4063	20.4609
Euclidean	SeqVec	0.5804	1.3425	30408.3304	-0.0675	**22.2295**	30.0684
	ESM	1.1336	1.5309	28876.8367	**-0.0748**	20.4887	31.0122
	TAPE	0.6024	1.2547	11401.2842	-0.0690	6.4066	**18.1227**
	ProtT5	**0.1073**	**0.2458**	**2487.9561**	-0.0653	1.4063	20.4609

Table 6: Comparison of geometric and topological metrics across different embeddings for the Breast Cancer dataset, grouped by the distance matrix.

Remark 1. *Because Ricci curvature, persistent homology, and fractal dimension are calculated directly on the embeddings rather than the complete pairwise distance matrix, they remain constant across the Euclidean and Poincaré distance matrices.*

6.2 Classification Results

The classification results on the PDB186 dataset are shown in Table 7. ProtT5 achieves the highest ROC-AUC (0.7968 with LR), followed by ESM2 (0.7453), while SeqVec shows poor performance (0.6173). We can see that across other metrics as well, ProtT5 consistently outperforms SeqVec. Table 8 shows the classification results for the Lung Cancer dataset. Here, ESM2 achieves the highest ROC-AUC (0.8568 with SVM), followed by ProtT5 (0.8311 with LR). SeqVec and TAPE consistently show the lowest performance, with only marginal

Embed.	Algo.	Acc. ↑	Prec. ↑	Recall ↑	F1 (Weig.) ↑	F1 (Macro) ↑	ROC AUC ↑	Train Time (sec.) ↓
SeqVec	SVM	0.5786	0.6045	0.5786	0.5720	0.5755	0.5917	1692.2908
	NB	0.5036	0.5538	0.5036	0.3878	0.3836	0.5014	132.7588
	MLP	0.5607	0.5934	0.5607	0.5088	0.4911	0.5330	845.1396
	KNN	0.5107	0.3741	0.5107	0.3815	0.3812	0.5135	1808.8480
	RF	0.5857	0.5961	0.5857	0.5857	0.5856	0.5908	100.2860
	LR	0.6179	0.6216	0.6179	0.6184	0.6159	0.6173	1024.6331
	DT	0.5536	0.5550	0.5536	0.5531	0.5494	0.5507	277.4930
ESM2	SVM	0.7321	0.7520	0.7321	0.7327	0.7319	0.7420	0.0138
	NB	0.6321	0.6359	0.6321	0.6325	0.6273	0.6289	0.0034
	MLP	0.7107	0.7274	0.7107	0.7116	0.7103	0.7190	0.7072
	KNN	0.6214	0.6531	0.6214	0.6159	0.6178	0.6354	0.0171
	RF	0.6964	0.7231	0.6964	0.6952	0.6951	0.7080	0.3342
	LR	0.7357	0.7561	0.7357	0.7361	0.7353	0.7453	0.0452
	DT	0.5857	0.5986	0.5857	0.5851	0.5821	0.5894	0.0674
TAPE	SVM	0.6286	0.6411	0.6286	0.6301	0.6269	0.6317	0.0185
	NB	0.6893	0.6965	0.6893	0.6897	0.6858	0.6896	**0.0031**
	MLP	0.6964	0.7170	0.6964	0.6950	0.6939	0.7032	33.7174
	KNN	0.6000	0.6134	0.6000	0.5991	0.5954	0.6023	0.5691
	RF	0.6714	0.6974	0.6714	0.6696	0.6684	0.6795	0.4725
	LR	0.6286	0.6398	0.6286	0.6301	0.6268	0.6315	0.1313
	DT	0.5179	0.5372	0.5179	0.5153	0.5149	0.5259	0.0619
ProtT5	SVM	0.7679	0.7958	0.7679	0.7681	0.7675	0.7813	0.0235
	NB	0.7250	0.7385	0.7250	0.7228	0.7181	0.7261	0.0035
	MLP	0.7464	0.7558	0.7464	0.7459	0.7415	0.7465	38.2089
	KNN	0.6821	0.6928	0.6821	0.6818	0.6784	0.6843	0.6559
	RF	0.7036	0.7291	0.7036	0.7029	0.7015	0.7136	0.3849
	LR	**0.7893**	**0.8029**	**0.7893**	**0.7903**	**0.7884**	**0.7968**	0.1281
	DT	0.5607	0.5826	0.5607	0.5585	0.5567	0.5687	0.0841

Table 7: Average classification results for different models and algorithms for **PDB186** dataset.

differences between them across other metrics. The classification results for the Breast Cancer dataset are shown in Table 9, and they reinforce the trend observed previously. ESM2 attains the highest ROC-AUC (0.8385), followed by ProtT5 at 0.8195, while SeqVec again shows the weakest performance (0.7597). Across all other metrics, SeqVec consistently lags behind by a noticeable margin, further highlighting the superiority of transformer-based embeddings for classification tasks.

6.3 Clustering Results

Table 10 presents the clustering results for the PDB186 dataset, while Table 11 and Table 12 show the results for the Lung Cancer and Breast Cancer datasets, respectively.

Embed.	Algo.	Acc. ↑	Prec. ↑	Recall ↑	F1 (Weig.) ↑	F1 (Macro) ↑	ROC AUC ↑	Train Time (sec.) ↓
SeqVec	SVM	0.9240	0.9212	0.9240	0.9202	0.6836	0.8212	29.3541
	NB	0.6347	0.7938	0.6347	0.6212	0.2585	0.6396	3.6623
	MLP	0.9048	0.9101	0.9048	0.9056	0.6513	0.8121	142.6836
	KNN	0.8812	0.8660	0.8812	0.8683	0.5436	0.7165	9.0200
	RF	0.9085	0.8955	0.9085	0.8977	0.6329	0.7649	7.2720
	LR	0.7882	0.8925	0.7882	0.8254	0.5433	0.8066	1005.2145
	DT	0.8531	0.8692	0.8531	0.8593	0.5348	0.7469	23.7610
ESM2	SVM	**0.9506**	**0.9474**	**0.9506**	**0.9475**	**0.7335**	**0.8568**	11.9609
	NB	0.6472	0.8906	0.6472	0.7109	0.5231	0.8066	0.6002
	MLP	0.9365	0.9453	0.9365	0.9389	0.7428	0.8698	9.4761
	KNN	0.9122	0.9259	0.9122	0.9144	0.6687	0.8300	1.2228
	RF	0.9380	0.9352	0.9380	0.9342	0.7144	0.8296	2.8020
	LR	0.7823	0.8957	0.7823	0.8171	0.5472	0.8243	156.7906
	DT	0.9011	0.9152	0.9011	0.9039	0.6469	0.8074	5.3667
TAPE	SVM	0.9166	0.9181	0.9166	0.9165	0.6177	0.7967	0.0394
	NB	0.9018	0.9204	0.9018	0.9082	0.6461	0.8306	**0.0095**
	MLP	0.9085	0.9152	0.9085	0.9102	0.6194	0.7943	1.7274
	KNN	0.9122	0.9143	0.9122	0.9107	0.5994	0.7771	0.0265
	RF	0.9144	0.9065	0.9144	0.9054	0.6336	0.7546	0.9389
	LR	0.9173	0.9252	0.9173	0.9200	0.6564	0.8221	0.7910
	DT	0.8657	0.8873	0.8657	0.8748	0.5586	0.7576	0.3775
ProtT5	SVM	0.9277	0.9313	0.9277	0.9277	0.6664	0.8290	0.0513
	NB	0.9151	0.9226	0.9151	0.9175	0.6461	0.8232	0.0125
	MLP	0.9196	0.9263	0.9196	0.9219	0.6375	0.8155	3.5043
	KNN	0.9188	0.9169	0.9188	0.9165	0.5826	0.7817	0.0649
	RF	0.9321	0.9293	0.9321	0.9258	0.6520	0.7997	1.0917
	LR	0.9277	0.9326	0.9277	0.9288	0.6724	0.8311	1.0877
	DT	0.8841	0.8977	0.8841	0.8887	0.5800	0.7800	0.4553

Table 8: Average classification results for different models and algorithms for **Lung Cancer** dataset.

For the PDB186 dataset, SeqVec shows an interesting contrast: it performs best on clustering metrics such as the Silhouette Coefficient and Davies–Bouldin Score in all clustering algorithms, yet it is the weakest model in the classification setting. ESM2, which ranked second in classification, shows more consistent behavior, particularly on the Calinski–Harabasz Score, where it performs well for Agglomerative and k-means clustering. This contrast highlights how models that excel in supervised classification do not necessarily produce the most cluster-friendly embedding spaces, and vice versa.

For the Lung Cancer dataset, ProtT5 performs better in all clustering metrics for Agglomerative and k-means. However, the k-modes findings are more in line with classification: ESM2, which is marginally more effective in supervised tasks, also receives higher k-modes scores, particularly on the Silhouette and

Embed.	Algo.	Acc. ↑	Prec. ↑	Recall ↑	F1 (Weig.) ↑	F1 (Macro) ↑	ROC AUC ↑	Train Time (sec.) ↓
SeqVec	SVM	0.9004	0.8995	0.9004	0.8977	0.5682	0.7597	1724.5
	NB	0.4996	0.7564	0.4996	0.5289	0.2087	0.5882	4.2771
	MLP	0.8716	0.8833	0.8716	0.8740	0.5218	0.7447	277.84
	KNN	0.7761	0.8375	0.7761	0.7969	0.4523	0.6859	175.88
	RF	0.8814	0.8649	0.8814	0.8671	0.5349	0.7082	11.531
	LR	0.7719	0.8811	0.7719	0.8164	0.5014	0.7528	1836.9
	DT	0.8133	0.8245	0.8133	0.8179	0.4515	0.6745	37.130
ESM2	SVM	**0.9333**	**0.9335**	**0.9333**	**0.9319**	**0.6843**	**0.8385**	8.0102
	NB	0.6681	0.8646	0.6681	0.7150	0.5013	0.7977	0.7043
	MLP	0.9067	0.9182	0.9067	0.9105	0.6353	0.8032	12.827
	KNN	0.9060	0.9089	0.9060	0.9047	0.6380	0.8268	1.1774
	RF	0.9326	0.9257	0.9326	0.9261	0.6589	0.7914	2.7038
	LR	0.7993	0.8796	0.7993	0.8239	0.5591	0.8177	154.53
	DT	0.8877	0.8889	0.8877	0.8871	0.6237	0.7948	3.7809
TAPE	SVM	0.8982	0.8968	0.8982	0.8962	0.5964	0.7813	0.0930
	NB	0.8575	0.8852	0.8575	0.8691	0.5760	0.7887	**0.0122**
	MLP	0.8884	0.8943	0.8884	0.8901	0.5825	0.7815	284.43
	KNN	0.8821	0.8716	0.8821	0.8745	0.5119	0.7345	1.1070
	RF	0.8870	0.8683	0.8870	0.8738	0.5615	0.7374	2.2642
	LR	0.9046	0.9082	0.9046	0.9030	0.6161	0.7961	1.9700
	DT	0.8386	0.8511	0.8386	0.8433	0.5123	0.7262	0.9652
ProtT5	SVM	0.8975	0.9011	0.8975	0.8973	0.6425	0.8086	0.0743
	NB	0.8842	0.8902	0.8842	0.8865	0.5744	0.7770	0.0142
	MLP	0.8961	0.9015	0.8961	0.8977	0.6509	0.8195	3.3378
	KNN	0.8933	0.8796	0.8933	0.8855	0.5230	0.7425	0.0496
	RF	0.9018	0.8892	0.9018	0.8934	0.5996	0.7651	1.3799
	LR	0.8947	0.8973	0.8947	0.8951	0.6131	0.8004	1.0668
	DT	0.8484	0.8535	0.8484	0.8497	0.5021	0.7278	0.6561

Table 9: Average classification results for different models and algorithms for **Breast Cancer** dataset.

Calinski–Harabasz scores. We can say that ProtT5 yields more cluster-friendly continuous embeddings, whereas ESM2 captures a structure better suited to categorical clustering.

We find a trend comparable to the Lung Cancer results in the Breast Cancer dataset. Although ESM2 outperforms ProtT5 in classification, it performs better in k-modes, particularly on the Silhouette and Calinski–Harabasz scores. ProtT5 has the best clustering performance for Agglomerative and k-means. This again exhibits a similar pattern: ProtT5 produces more cluster-friendly continuous embeddings, while ESM2 captures structure more closely aligned with categorical clustering and supervised classification tasks.

Algorithm	Embedding	Evaluation Metrics			Clustering Runtime ↓
		Silhouette Coefficient ↑	Calinski-Harabasz Score ↑	Davies-Bouldin Score ↓	
Agglomerative	Seqvec	**0.603**	9.872	**0.278**	**0.01**
	ESM2	0.230	44.368	1.348	0.02
	TAPE	0.136	12.068	3.665	**0.01**
	ProtT5	0.117	10.429	2.306	0.03
k-means	Seqvec	0.384	3.407	0.469	1.46
	ESM2	0.178	**51.234**	1.820	1.46
	TAPE	0.096	13.810	3.567	1.20
	ProtT5	0.055	11.117	3.963	1.78
k-modes	Seqvec	0.312	2.545	0.543	0.73
	ESM2	0.161	2.118	0.657	3.58
	TAPE	0.033	1.167	0.895	2.32
	ProtT5	-0.075	0.698	1.161	2.73

Table 10: Internal clustering quality metrics for k-means and k-modes on different embeddings on the PDB186. Best values are shown in bold.

Algorithm	Embedding	Evaluation Metrics			Clustering Runtime ↓
		Silhouette Coefficient ↑	Calinski-Harabasz Score ↑	Davies-Bouldin Score ↓	
Agglomerative	Seqvec	-0.002	36.713	3.792	**0.22**
	ESM2	0.048	75.235	3.012	9.67
	TAPE	0.082	80.764	2.709	0.23
	ProtT5	0.177	88.294	2.030	0.25
k-means	Seqvec	0.026	38.255	3.922	2.29
	ESM2	0.070	79.774	3.370	5.95
	TAPE	0.118	**98.614**	2.527	2.54
	ProtT5	**0.197**	94.129	1.947	1.58
k-modes	Seqvec	-0.038	0.947	1.023	16.12
	ESM2	0.035	44.883	6.223	1642.4
	TAPE	-0.168	0.837	1.087	18.40
	ProtT5	-0.162	1.388	**1.023**	22.34

Table 11: Internal clustering quality metrics for k-means and k-modes on different embeddings on the Lung Cancer. Best values are shown in bold.

6.4 Overall Results - Observations

Across all three datasets, we observe a pattern indicating the importance of geometry for downstream performance. The classification task aligns with a strong Poincaré hierarchy, in which ESM2 exhibits a better hyperbolic structure and consistently performs better on supervised tasks. Clustering aligns with Euclidean space, and ProtT5 exhibits the most separable Euclidean structure and gives better performance with Agglomerative and k-means clustering.

Algorithm	Embedding	Evaluation Metrics			Clustering Runtime ↓
		Silhouette Coefficient ↑	Calinski-Harabasz Score ↑	Davies-Bouldin Score ↓	
Agglomerative	Seqvec	-0.002	38.110	3.868	0.30
	ESM2	0.048	78.754	3.064	10.75
	TAPE	0.096	87.078	2.826	0.17
	ProtT5	**0.206**	100.927	2.003	**0.14**
k-means	Seqvec	-0.004	37.709	3.765	2.41
	ESM2	0.072	84.914	3.380	5.71
	TAPE	0.122	**109.023**	2.605	1.56
	ProtT5	0.190	106.776	2.021	2.32
k-modes	Seqvec	-0.093	0.910	1.078	18.29
	ESM2	0.037	47.333	5.461	2277.34
	TAPE	-0.116	1.035	0.984	8.96
	ProtT5	-0.169	2.356	**0.957**	9.50

Table 12: Internal clustering quality metrics for k-means and k-modes on different embeddings on the Breast Cancer. Best values are shown in bold.

Although it occasionally clusters well, SeqVec, the worst in classification, also has the most fragmented geometry, which explains its inconsistent behavior. Although TAPE exhibits a strong hyperbolic hierarchy, it does not translate into downstream performance, indicating that the hyperbolic structure alone is insufficient. In general, ESM2 excels at classification, ProtT5 at clustering, and SeqVec and TAPE exhibit weaker or less consistent geometric patterns.

7 Conclusion

In this paper, we present a geometric framework for evaluating LLM embeddings using δ-hyperbolicity, ultrametricity, Neighbor Joining, and complementary measures such as Ricci curvature, persistent homology, and fractal dimension. We found that the embedding geometry aligns with the downstream performance. We found that ESM2's hyperbolic structure supports strong classification performance, that ProtT5's Euclidean separability explains its superior clustering, and that weaker models like SeqVec show fragmented geometric patterns consistent with lower task performance. The results highlight that no

single embedding is universally optimal. Each model excels in different geometric regimes. Each model is better suited to a specific task, and this structural analysis can pave the way towards guided, informed designs and the selection of LLMs. Our analysis of synthetic spaces further confirms how hyperbolicity varies across geometric settings, emphasizing the importance of understanding embedding structure. All things considered, geometric metrics provide a useful perspective for choosing and fine-tuning LLM embeddings. Future work can extend this framework to broader datasets and explore integrating geometric insights into model design and training.

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